

1. (a) Analog signal $x_a(t) = 4 + 2 \cos(150\pi t + \pi/3) + 4 \sin(350\pi t)$ is sampled at $F_s = 200$ samples/sec to obtain the DT signal $x(n)$. Determine $x(n)$ & DTFT $X(e^{j\omega})$.

(b) Consider that the sampled signal $x(n)$ obtained in part (a) is applied to an ideal D/A converter [i.e., an ideal interpolator] to obtain analog signal $y_a(t)$. The ideal D/A converter is also operating at $F_s = 200$ samples/sec. Obtain the reconstructed signal $y_a(t)$ and determine whether the sampling or reconstruction opsⁿ resulted in any aliasing.

2. Given $x_1(n) = \{1, 2, 2\}$ and $x_2(n) = \{1, 2, 3, 4\}$. Compute 4-pt circular convolution: $x_1(n) \textcircled{4} x_2(n)$.

3. Using Parseval's relation for DTFT evaluate

(a) $\int_0^\pi \frac{4}{5 + 4 \cos \omega} d\omega$

(b) $\int_0^\pi \frac{1}{3.25 - 3 \cos \omega} d\omega.$

4. Let $x(n)$ be real with F.T. $X(e^{j\omega})$. Find $y(n)$ such that its F.T. is $Y(e^{j\omega}) = X(e^{j3\omega})$.

5. Even samples of 11 point DFT of length 11 real sequence are $X(0) = 4$, $X(2) = -1 + j3$, $X(4) = 2 + j5$, $X(6) = 9 - 6j$, $X(8) = -5 - j8$, $X(10) = \sqrt{3} - j2$. Find Missing samples.

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